

29	A nuclear fusion reaction is as follows: ${}^7_3\text{Li} + {}^1_1\text{H} \rightarrow 2 {}^4_2\text{He}$ The masses of the nuclei are as follow: ${}^7_3\text{Li}$: $7.018u$ ${}^1_1\text{H}$: $1.008u$ ${}^4_2\text{He}$: $4.004u$ What is the amount of energy released in this reaction?							
	A	$9.0 \times 10^{-21} \text{ J}$	B	$2.7 \times 10^{-12} \text{ J}$	C	$6.0 \times 10^{-10} \text{ J}$	D	$1.6 \times 10^{15} \text{ J}$
L2	Answer: B Energy released per fusion reaction = (mass of reactants – mass of products) (c^2) = $[(7.018 + 1.008) - 2(4.004)](1.66 \times 10^{-27}) (3.00 \times 10^8)^2$ = $2.6892 \times 10^{-12} \text{ J}$ = $2.7 \times 10^{-12} \text{ J}$							

30	${}^{218}\text{Po}$ decays to ${}^{214}\text{Pb}$ in two pathways as shown in the figure below.
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